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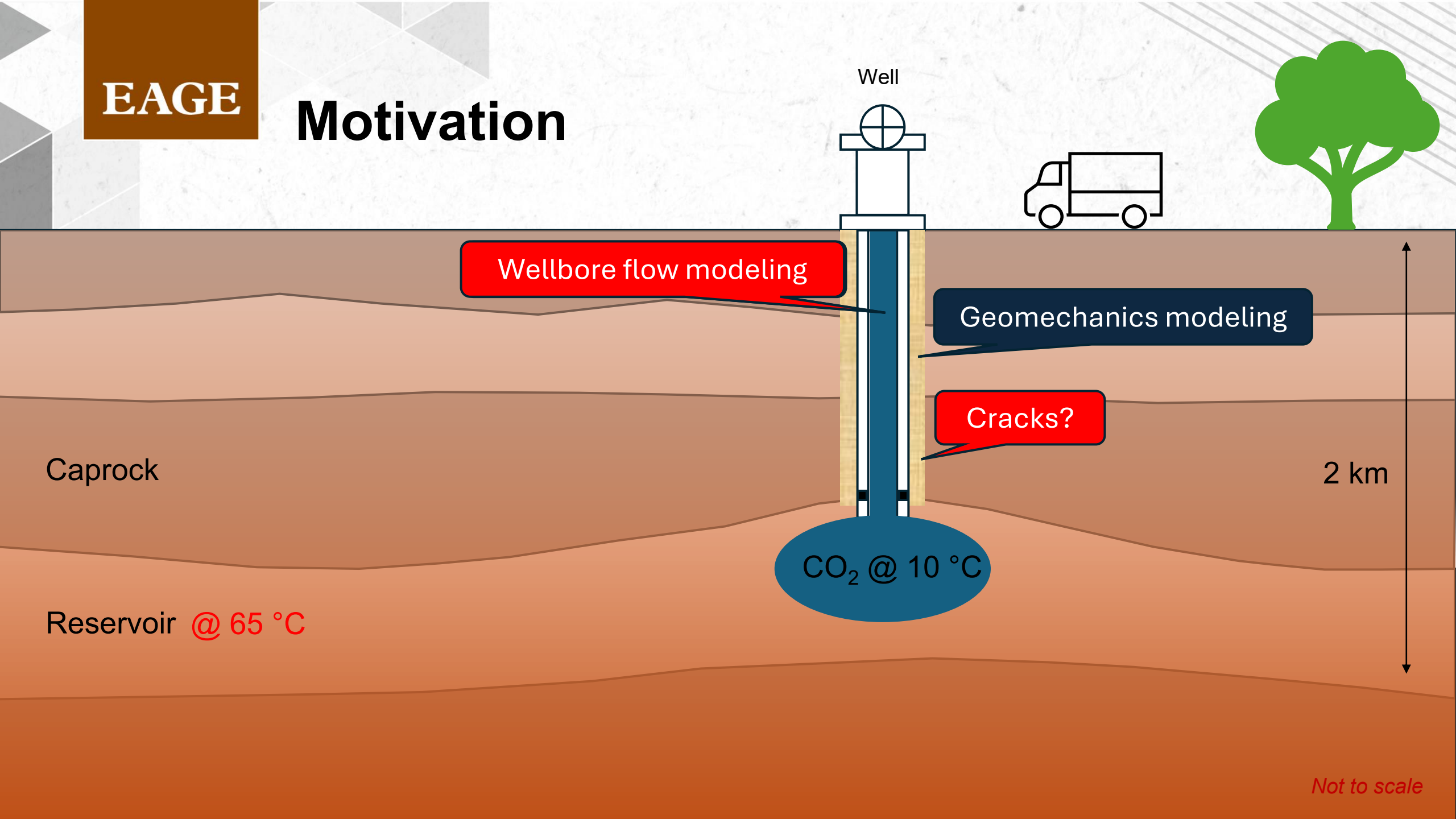
Open-source modeling of coupled wellbore-reservoir CO₂ injection using MRST

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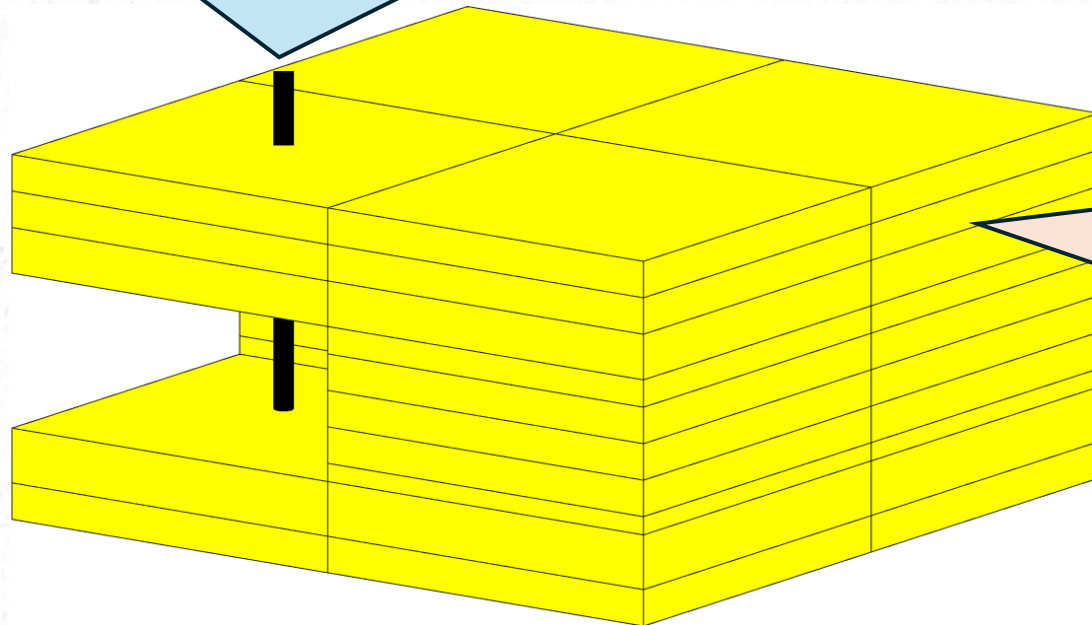
G E U S

Motivation



Problem formulation

- 1D multi-segment well
- One or two phases: water and CO₂ (gaseous, liquid or supercritical)
- Pressure drop equation, mass and energy conservation
- Heat conduction through annulus and cement



- 3D reservoir grid
- Two phases: water + CO₂ (liquid or supercritical)
- Darcy flow, mass and energy conservation (advection + conduction)

Reservoir flow

$$\phi \frac{\partial(\rho_j S_j)}{\partial t} + \nabla \cdot (\rho_j \mathbf{u}_j) = q_j, \quad j = w, g \quad (g = \text{scCO}_2)$$

$$\mathbf{u}_j = -\frac{\mathbf{K} k_{rj}(S_j)}{\mu_j} (\nabla p_j - \rho_j g \nabla z)$$

$$\begin{aligned} & \frac{\partial}{\partial t} \left[(1 - \phi) \rho_R C_{pR} T + \phi \sum_j \rho_j S_j u_j \right] \\ & + \nabla \cdot \left(\sum_j \rho_j h_j \mathbf{u}_j \right) \\ & = \nabla \cdot (\lambda_{\text{eff}} \nabla T) + Q_T \end{aligned}$$

- Two-phase immiscible Darcy flow
- No interphase mass transfer
- Energy conservation: rock+fluid accumulation, advection by both phases, conduction
- Local thermal equilibrium

Wellbore flow

$$\frac{\partial}{\partial t} (A x_j \rho_m) + \frac{\partial}{\partial s} (A x_j \rho_m v_m) = q_j$$

$$\frac{\partial p}{\partial z} - \rho_m g + \frac{2f \rho_m v_m |v_m|}{D} = 0$$

$$\frac{\partial}{\partial t} (A \rho_m E_m) + \frac{\partial}{\partial s} (A v (\rho_m E_m + p)) = -q_s + \varepsilon$$

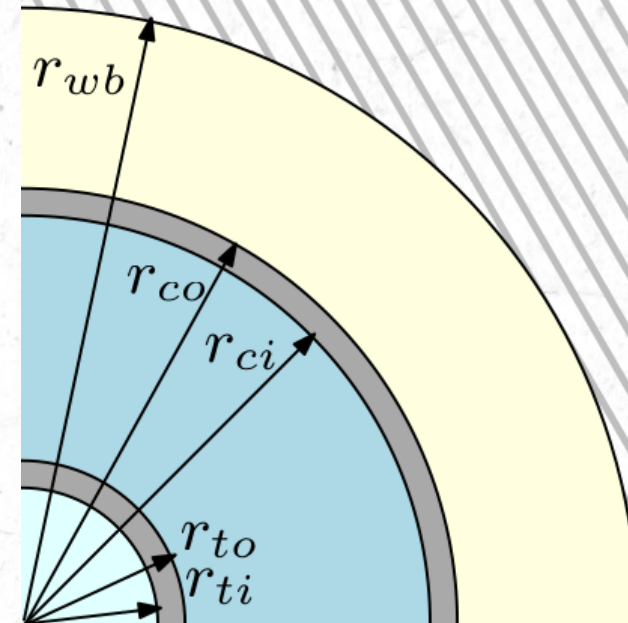
$$E_m = h_m - \frac{p}{\rho_m} + \frac{v^2}{2} - gz$$

- 1D area-averaged two-phase flow
- Phases have the same mixture velocity (no slip) and same pressure
- Pressure drop equation with gravity and friction terms (acceleration neglected)
- Mixture energy equation, formulated in mixture enthalpy $h_m = x_w h_w + x_g h_g$

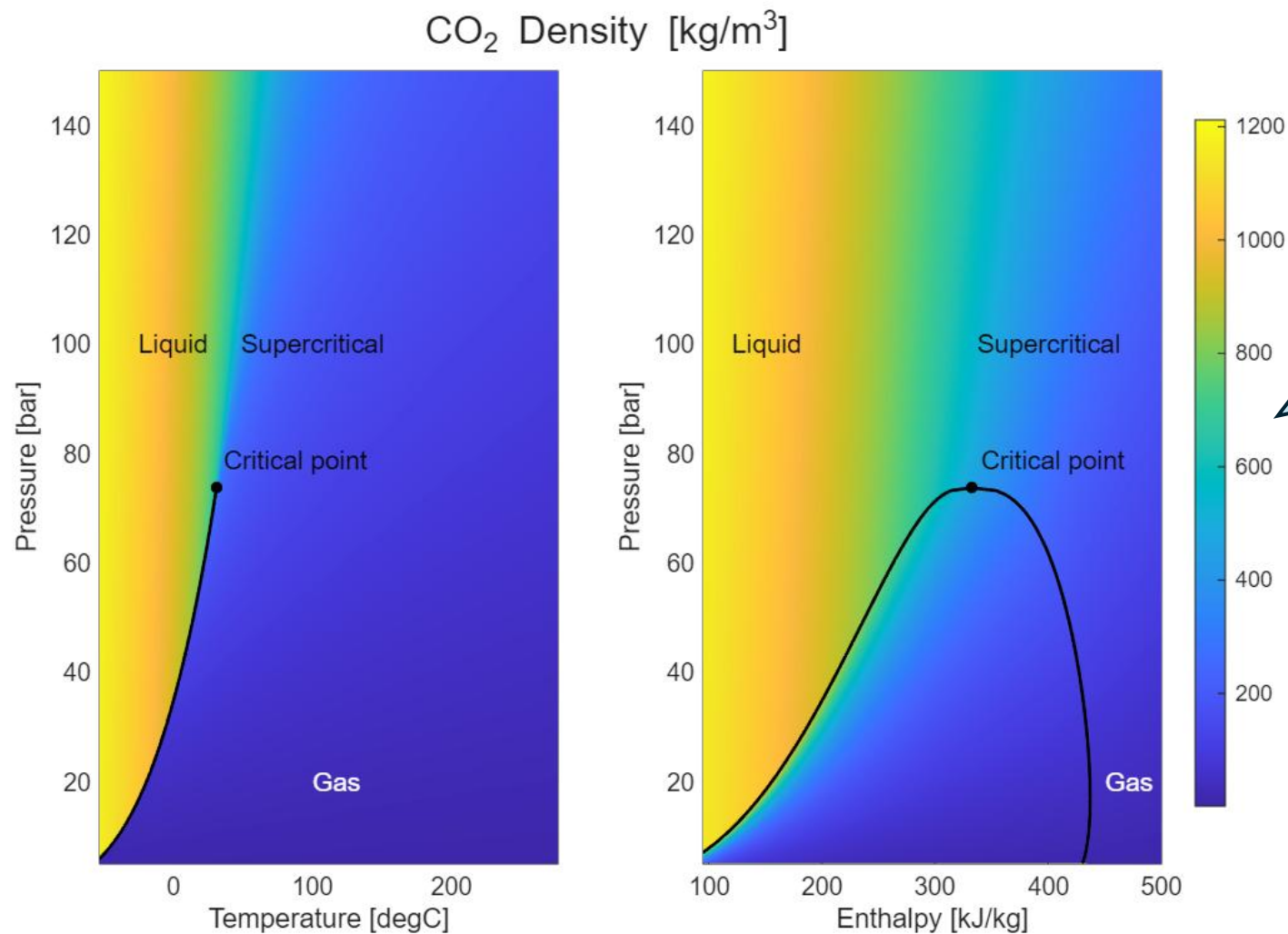
Formation heat loss

$$q_s = -\frac{2\pi U k_e}{k_e + 0.5\pi U T_D(t_D)} (T_e - T)$$

- Hasan–Kabir transient model with U built from tubing/annulus/casing/cement thermal conductivities, and $T_D(t_D)$ an analytical approximation to transient radial conduction
- Neglect convection in annulus



Thermodynamic modeling of CO₂



In the wellbore equations, use the pressure-enthalpy formulation to avoid jumps in density and viscosity across the saturation line

Reservoir-wellbore coupling

- Peaceman model to connect wellbore perforations to reservoir cells
- Implicit coupling: reservoir + wellbore unknowns assembled into one Jacobian, solved together by Newton–Raphson every timestep
- Primary unknowns
 - Reservoir: water pressure, gas saturation, temperature
 - Wellbore: node pressure, node phase mass fractions, segment mixture mass flux, node mixture enthalpy, node temperature

Why MRST

- Extensible open-source + modular class structure
- Automatic-differentiation (AD) framework gives exact Jacobians without hand-coded derivatives
- Multi-segment well (MSW) model available – need to add an energy equation
- Geothermal MRST modules show how to include temperature
- AD-compatible tabulated Span–Wagner CO₂ properties

Getting the job done – with AI – Sonnet

Sections in main.m	What and how it was built
Geometry, Rock & Fluid props, Init, Schedule, Visualization	• Re-used MRST examples • Done analogous to other MRST classes
Reservoir model	✓ TwoPhaseWaterGasThermalModel: T is primary variable. Single MSW.
Well model (MSW)	✓ Well geometry (nodes, segments, props) ✓ TwoPhaseThermalMultisegmentWell: T is primary variable.

TOTAL: Several sessions each few hours, single-digit \$\$

Object-oriented bot

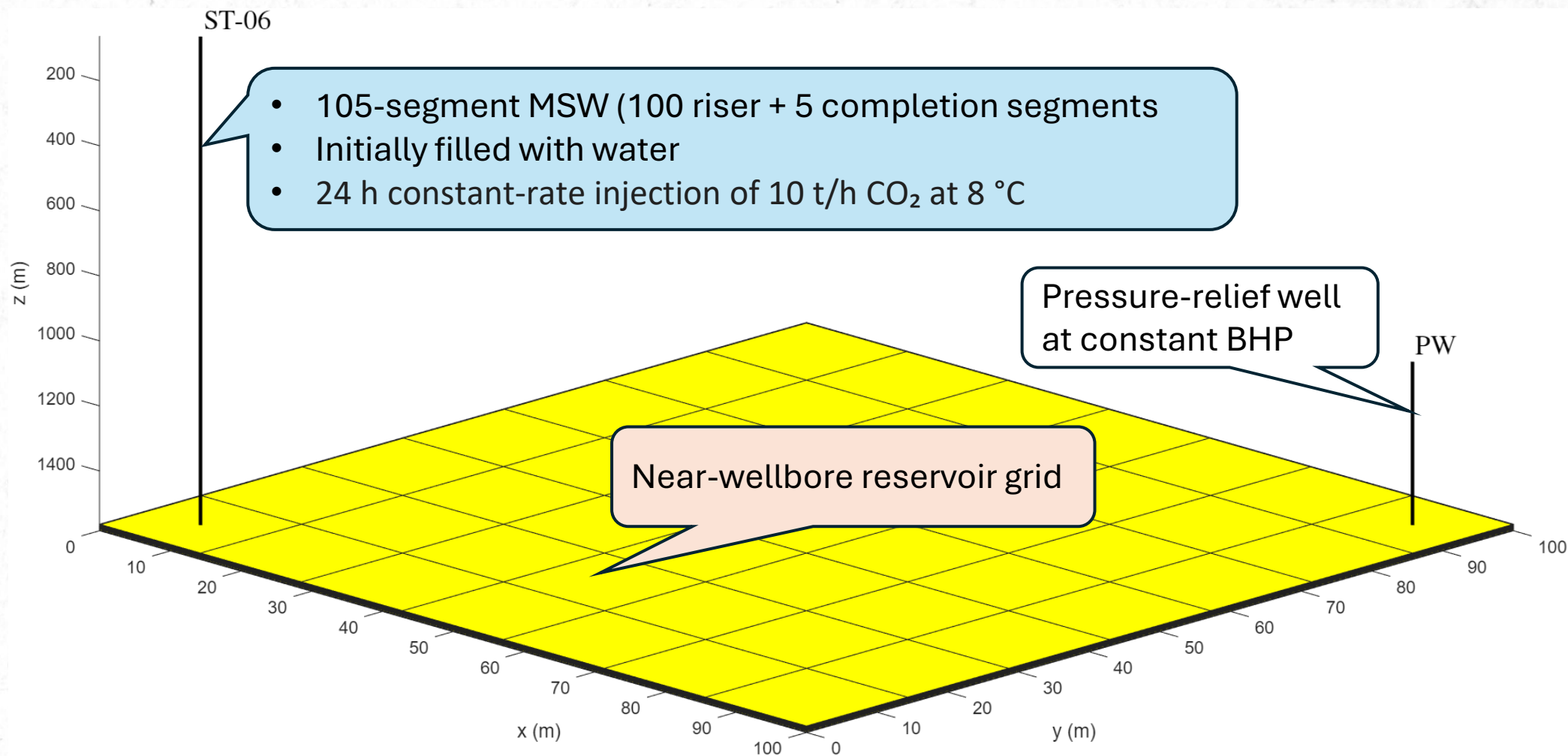
- One-to-one correspondence between plain description and MRST (`cartGrid`, `initSimpleADIFluid`, ...)
- Introduce JSON "bricks": atomic entities which should not be zoomed in any further (e.g. `SimpleWell`)
- A dictionary of text descriptions and corresponding bricks can be used as RAG to build new functionality
- Built a chatbot to create new bricks
- ✗ No proactive brick identification: hallucinated call patterns
- ✗ Worked for simple cases, but not for complex ones (e.g. enthalpy formulation in MSW)

TOTAL: 40 Claude sessions, approx. 100 hours, 193 USD

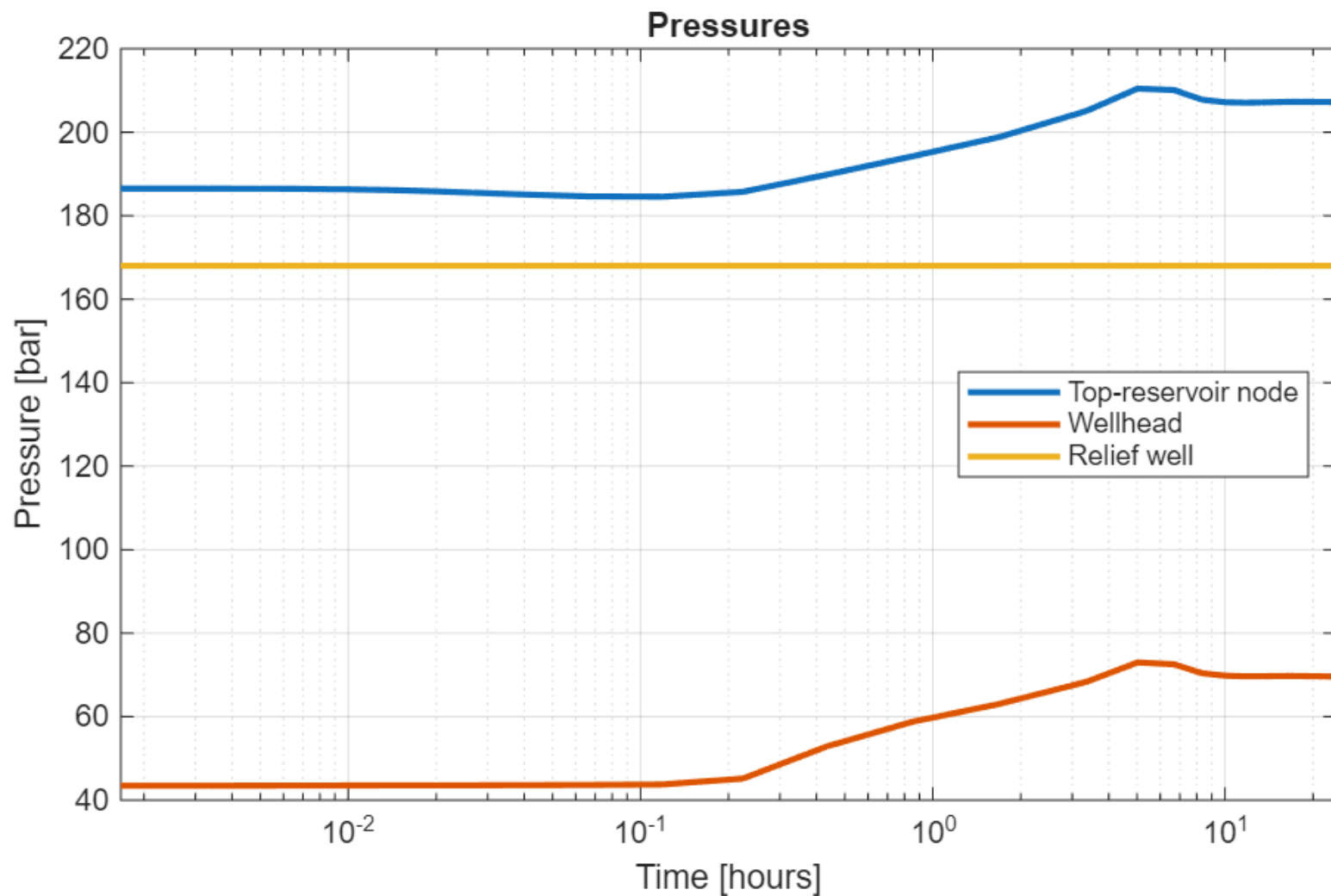
Getting the job done – with AI – Fable

Sections in main.m	What and how it was built
Well model (MSW)	✓ TwoPhaseThermalMultisegmentWell: added hydrostatic initialization ✓ TwoPhaseEnthalpyMultisegmentWell • Solves implicit equation to get T from the mixture enthalpy formulation using bisections, then evaluates the AD gradients with finite differences • Formulated equations in terms of (p, h) • For tabulated CO₂ properties, built an interpolation in (p, h) instead of (p, T) to get convergence near phase boundaries • Scaled the energy equation to improve convergence • Re-used a formulation for heat loss into formation from my old codes • Sometimes uses hardcoded values
TOTAL: 5 Claude sessions, approx. 17 hours, 246 USD	

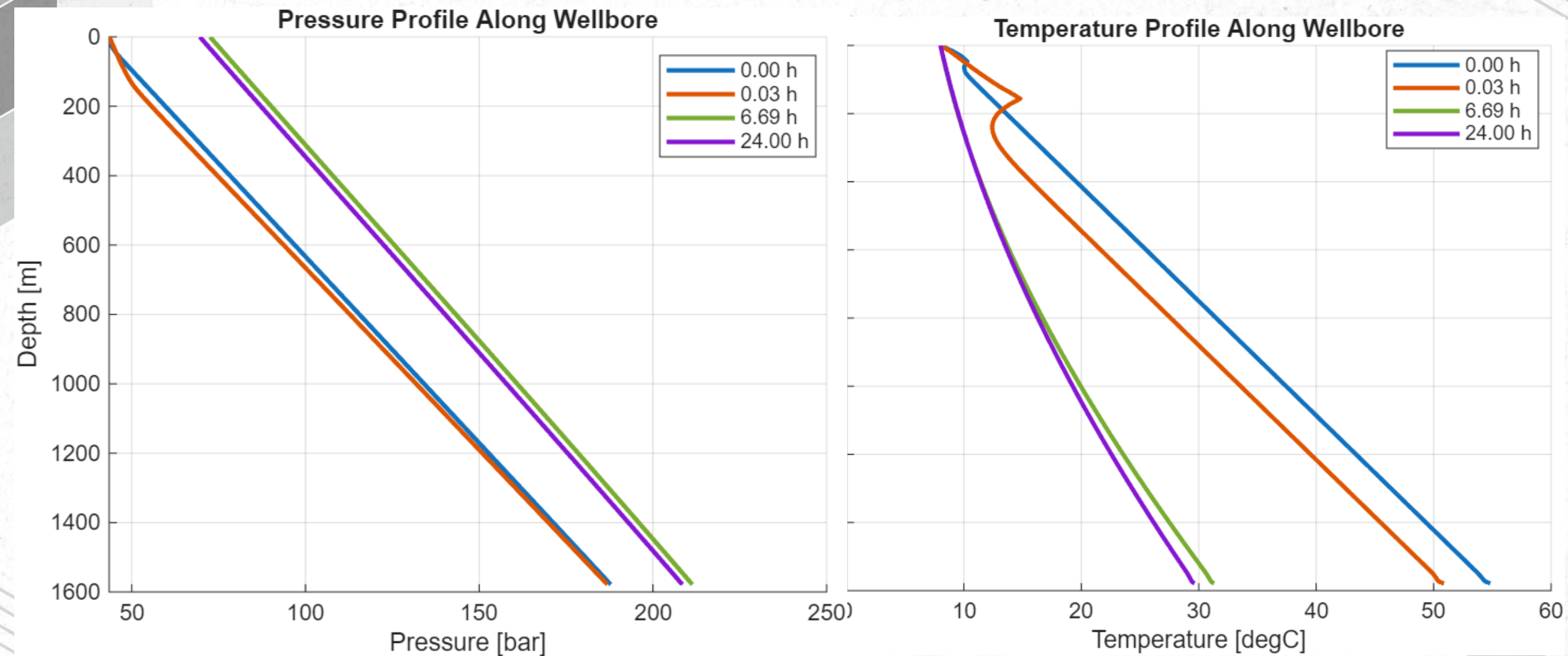
Case study



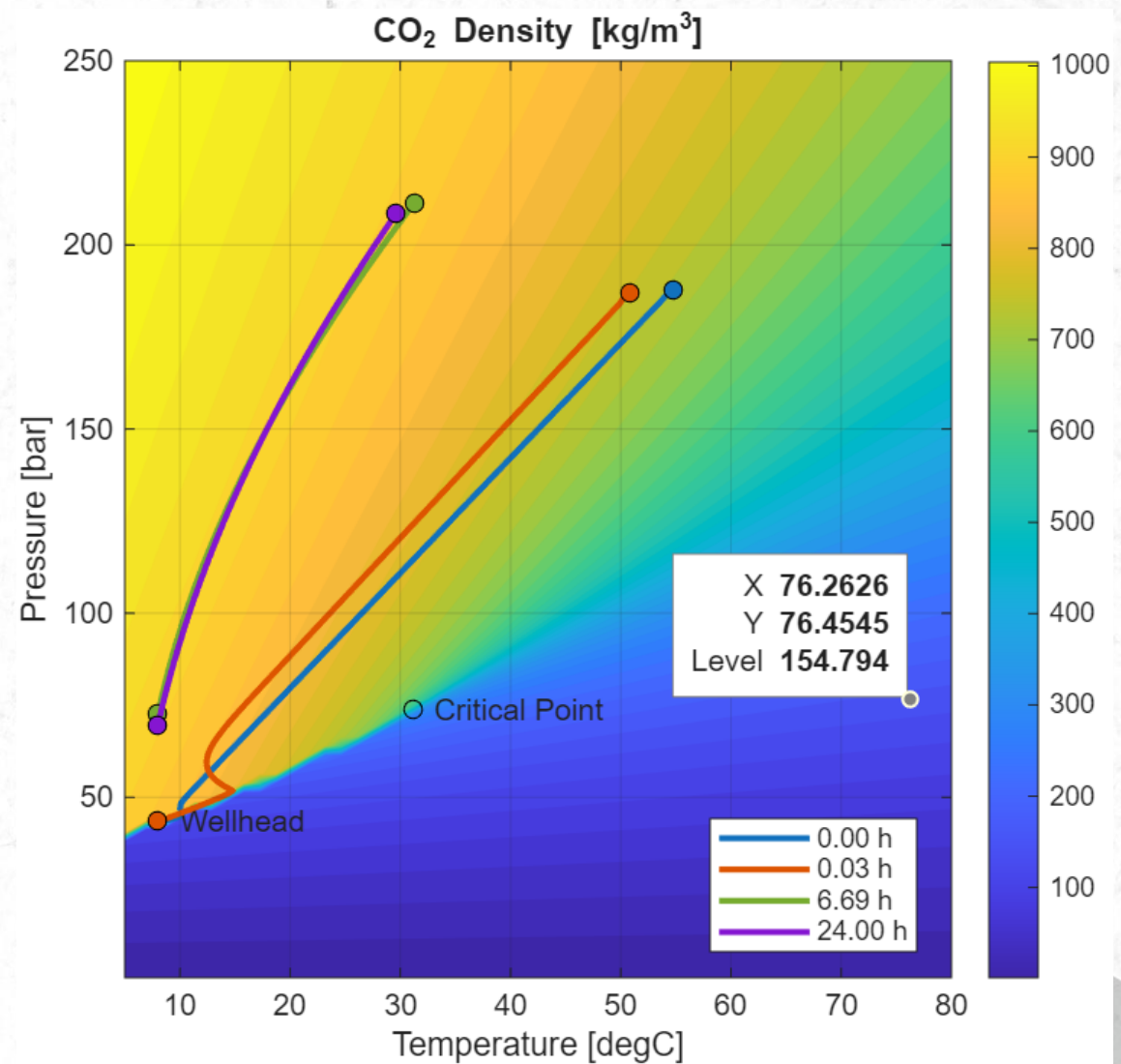
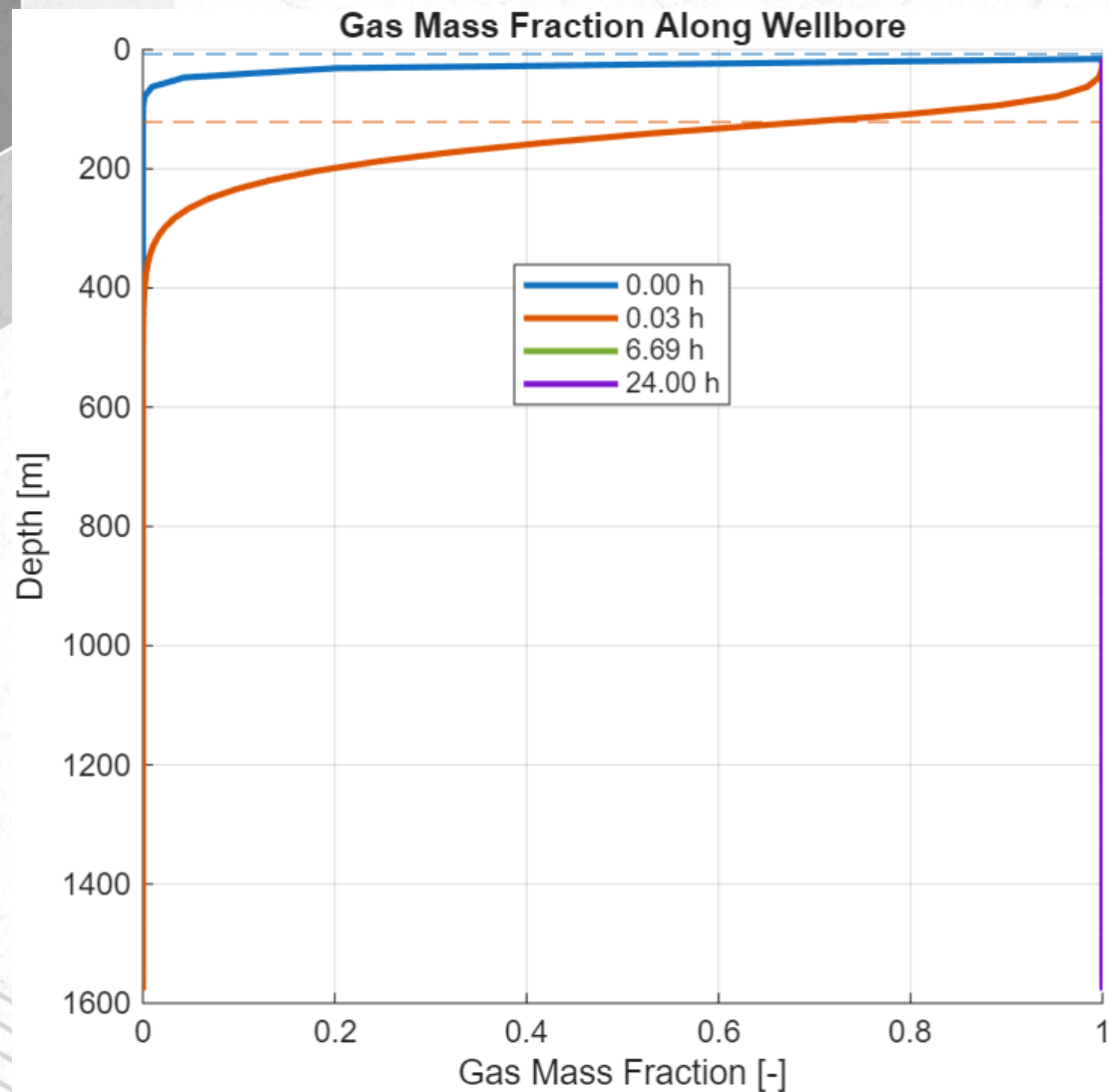
Results: pressure evolution



P & T profiles along the well



Wellbore dynamics & Phase behavior



Conclusions

- Open-source coupled wellbore–reservoir simulator for transient, non-isothermal CO₂ injection, implemented in MRST
- Enthalpy formulation is key to robust Newton convergence through CO₂ phase transitions
- Applied to Stenlille: quantifies the transient thermal loading (≈ 22 K contrast) that drives near-wellbore thermal stresses — the direct input for well-integrity assessment

Acknowledgements / Thank You / Questions

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- Questions?